

Brock Biology of Microorganisms, 15e (Madigan et al.)
Chapter 30 Person-to-Person Bacterial and Viral Diseases

30.1 Multiple Choice Questions

1) Which of the following microbial groups is most resistant to desiccation due to their cell wall?

- A) gram-positive
- B) gram-negative
- C) mycoplasmas
- D) viruses

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.1

2) The causative agent of streptococcal pharyngitis, also known as "strep throat," is

- A) *Streptococcus pyogenes*.
- B) *Streptococcus pneumoniae*.
- C) *Streptococcus mutans*.
- D) *Streptococcus lactis*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

3) Impetigo is due to an infection caused by

- A) *Staphylococcus aureus* but not *Streptococcus pyogenes*.
- B) *Streptococcus pyogenes* but not *Staphylococcus aureus*.
- C) either *Staphylococcus aureus* or *Streptococcus pyogenes*.
- D) neither *Staphylococcus aureus* nor *Streptococcus pyogenes*.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

4) Rheumatic fever is

- A) an autoimmune disease.
- B) a poststaphylococcal disease.
- C) triggered by cell-surface antigens on *Staphylococcus aureus* cells that are similar to heart valve and joint antigens.
- D) caused by human papillomavirus.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

5) Influenza A virus can be identified by which of the following surface glycoproteins?

- A) HA
- B) NA
- C) both HA and NA
- D) M protein

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.8

6) Encapsulated strains of *Streptococcus pneumoniae* are particularly pathogenic, because they

- A) produce exotoxin A.
- B) produce exotoxin B.
- C) are potentially very invasive.
- D) produce M protein surface antigens.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

7) A frequent symptom of the common cold is

- A) nasal discharge.
- B) fever.
- C) headache.
- D) diarrhea.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.7

8) Which of the following is a cure for mumps?

- A) a vaccine
- B) an antibiotic
- C) host antibodies
- D) an antiviral

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.6

9) Once a person has been infected with *Neisseria gonorrhoeae*, the individual

- A) acquires transient immunity, which prevents subsequent infection for about a year in most cases.
- B) can be reinfected immediately because of the diversity of strains and frequency of mutation.
- C) develops long-term immunity lasting for multiple years.
- D) should be treated using antiserum.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.13

10) The Spanish flu of 1918 was caused by an _____ influenza virus.

- A) H2N1
- B) H5N1
- C) H1N1
- D) H2N2

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.8

11) Which of the following statements about chlamydial infections is TRUE?

- A) Chlamydial diseases are not reportable diseases.
- B) Chlamydial diseases are less prevalent than gonorrhea.
- C) Chlamydial diseases can lead to acute complications.
- D) All of the statements are true.

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.14

12) The CD4 molecule is

- A) an HIV surface glycoprotein.
- B) a cell-surface protein that allows HIV to infect a cell.
- C) found only on T-helper cells.
- D) an HIV surface glycoprotein that is found only on T-helper cells.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

13) Which of the following drugs, chemotherapeutic agents against HIV, inhibits reverse transcriptase?

- A) ampligen
- B) azidothymidine
- C) alpha interferon
- D) soluble CD4

Answer: B

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 30.15

14) Which statement is TRUE about *Streptococcus pneumoniae* and the lung infection it causes?

- A) *Streptococcus pneumoniae* can spread from the focus of infection as a bacteremia.
- B) Penicillin and erythromycin are the "last chance" drugs for treatment of streptococcal pneumonia.
- C) Infection with any strain of encapsulated virulent *Streptococcus pneumoniae* provides long-lasting immunity to all encapsulated virulent strains.
- D) Streptococcal pneumonia is not as serious as most people think; if left untreated, it usually runs its course in about a week to ten days.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

15) Which of the following is/are used to identify methicillin-resistant *Staphylococcus aureus*?

- A) selective media only
- B) differential media only
- C) PCR only
- D) selective media, differential media, and PCR can all be used

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

16) Which of the following is the causal agent of gastric ulcers?

- A) *Staphylococcus aureus*
- B) *Escherichia coli* O157:H7
- C) *Helicobacter pylori*
- D) *Vibrio cholerae*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.10

17) The diphtheria component of the DTaP vaccine is a(n)

- A) toxin.
- B) toxoid.
- C) antitoxin.
- D) antitoxoid.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

18) Which of the following is NOT a staphylococcal disease?

- A) acne
- B) boils
- C) strep throat
- D) meningitis

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

19) Isonicotinic acid hydrazide (INH) is an antibiotic used in the treatment of tuberculosis. It affects the synthesis of

- A) mycolic acid.
- B) linoleic acid.
- C) globular proteins of the cell membrane.
- D) DNA transcription.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

20) How is the syphilis spirochete initially transmitted?

- A) penetration of unbroken skin anywhere on the body
- B) through the eyes and mucous membranes
- C) through ingestion
- D) through tiny breaks in the skin epithelial layer

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

21) The single most important physical sign of primary syphilis is

- A) the chancre.
- B) a skin rash.
- C) paralysis.
- D) an acute eye infection.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

22) Tuberculoid Hansen's disease is caused by the pathogen

- A) *Listeria monocytogenes*.
- B) *Mycobacterium leprae*.
- C) *Mycobacterium tuberculosis*.
- D) either *Mycobacterium leprae* or *Mycobacterium tuberculosis*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

23) Of the following, which is the most common AIDS-associated opportunistic disease?

- A) *Pneumocystis jirovecii* pneumonia
- B) cryptosporidiosis
- C) histoplasmosis
- D) cryptococcosis

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

24) Which of the following statements is TRUE about HIV infections?

- A) The viral nucleocapsid of the virus enters the host cell when the viral and host membranes fuse.
- B) T-helper cells are greatly reduced in number.
- C) The cDNA can integrate into the host chromosome.
- D) All of these statements are true.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.15

25) Serious acute cases of diphtheria are treated with antibiotics and

- A) diphtheria antitoxin.
- B) diphtheria toxin.
- C) diphtheria toxoid.
- D) diphtheria antitoxin, diphtheria toxin, and diphtheria toxoid.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

26) The unique antigenic determinants of the different serotypes of *Streptococcus pneumoniae* are found

- A) on the cell-surface M proteins.
- B) on the capsular polysaccharides.
- C) in the bacterial cell wall.
- D) in the inclusions in the periplasmic space.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

27) The DTaP vaccine contains

- A) purified bacterial DNA.
- B) bacterial endotoxin derivatives.
- C) heat-killed *Bordetella pertussis* cells.
- D) diphtheria toxoid.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

28) Which of the following are group A *Streptococcus* (GAS)?

- A) *Streptococcus agalactiae*
- B) *Streptococcus pneumonia*
- C) *Streptococcus pyogenes*
- D) *Streptococcus viridans*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

29) Erythromycin is applied to the eyes of newborns particularly to prevent

- A) syphilis.
- B) gonorrhea.
- C) a chlamydial infection.
- D) ocular herpes.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

30) Which of the following does NOT apply to bacterial pathogens affecting the respiratory tract?

- A) Humans are the only reservoir for most bacterial respiratory pathogens.
- B) Most are normally transmitted person to person.
- C) Some often initiate secondary infections that can be life threatening.
- D) Most respiratory bacterial pathogens do not respond to antibiotic therapy.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.1

31) The presence of gram-positive diplococci in a sputum sample most likely indicates an infection of

- A) *Streptococcus pyogenes*.
- B) *Streptococcus pneumoniae*.
- C) *Corynebacterium diphtheriae*.
- D) *Bordetella pertussis*.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.5

32) Whooping cough is frequently observed in

- A) elderly residents of nursing homes.
- B) individuals with compromised immunity.
- C) children under 6 months of age.
- D) health care providers.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

33) Which disease is the global top infectious disease killer, causing over 1.5 million deaths per year worldwide?

- A) diphtheria
- B) whooping cough
- C) pneumococcal pneumonia
- D) tuberculosis

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

34) Which of the following is TRUE of tuberculosis?

- A) the causative agent can remain dormant in macrophages
- B) it used to be very common, but is now rare
- C) it is caused by a virus
- D) very short antibiotic treatments of just a few days are usually sufficient for treatment

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

35) The most prevalent human infectious diseases are caused by

- A) bacteria.
- B) viruses.
- C) fungi.
- D) protozoans.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.6

36) The most common infectious disease is

- A) gonorrhea.
- B) influenza.
- C) hepatitis.
- D) the common cold.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.7

37) Pus formation in skin lesions caused by *Staphylococcus aureus* infections is in part due to production of

- A) coagulase.
- B) hemolysins.
- C) leukocidin.
- D) catalase.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

38) "Bloodborne hepatitis" is caused by infection with hepatitis _____ virus.

- A) A
- B) B
- C) C
- D) E

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.11

39) Which organism causes the highest number of sexually transmitted diseases each year?

- A) *Neisseria gonorrhoeae*
- B) *Treponema pallidum*
- C) *Chlamydia trachomatis*
- D) *Trichomonas vaginalis*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

40) Hepatitis B and C are transmitted via the

- A) oral route.
- B) parenteral route.
- C) dermal route.
- D) ophthalmic route.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.11

41) A positive tuberculin test means a patient

- A) may have active disease.
- B) may have been previously exposed to *Mycobacterium tuberculosis*.
- C) may have a current inapparent infection.
- D) may have active disease, been previously exposed to *Mycobacterium tuberculosis*, or may have a current inapparent infection.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.4

42) The inflammatory response to *Corynebacterium diphtheriae* infection results in the production of a lesion called a

- A) pseudomembrane.
- B) chancre.
- C) lesion.
- D) pustule.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

43) Another name for pertussis is

- A) the kissing disease.
- B) shingles.
- C) whooping cough.
- D) yellow fever.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

44) The transmission of _____ has been greatly reduced by the pasteurization of milk.

- A) *Mycobacterium marinum*
- B) *Mycobacterium ulcerans*
- C) *Mycobacterium hansen*
- D) *Mycobacterium bovis*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

45) *Staphylococcus aureus* produces _____, an enzyme that results in the accumulation of fibrin around the bacterial cells.

- A) coagulase
- B) collagenase
- C) lipase
- D) M protein

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

46) Which of the following is NOT a sign of scarlet fever?

- A) fever
- B) rash
- C) sore throat
- D) impetigo

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

47) Lymphogranuloma venereum is a sexually transmitted disease caused by distinct strains of _____.

- A) *Chlamydia trachomatis*
- B) herpes simplex 2 virus (HSV-2)
- C) *Neisseria gonorrhoeae*
- D) *Haemophilus ducreyi*

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

48) Nongonococcal urethritis may be caused by the protozoan _____.

- A) *Chlamydia trachomatis*
- B) *Neisseria gonorrhoeae*
- C) *Haemophilus ducreyi*
- D) *Trichomonas vaginalis*

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

49) Several strains of _____ cause genital warts and cervical cancer.

- A) human papillomavirus
- B) *Neisseria gonorrhoeae*
- C) *Chlamydia trachomatis*
- D) HIV

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

50) The HPV vaccine has been recommended for

- A) females ages 11-26 only.
- B) females ages 26-45 only.
- C) males ages 11-26 only.
- D) males and females ages 11-26.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

51) Individuals who have an active case of tuberculosis may spread the disease simply by _____ uninfected individuals.

- A) coughing near
- B) shaking hands with
- C) hugging
- D) sexual contact with

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

52) Transmission of *Mycobacterium leprae* involves

- A) direct contact.
- B) respiratory transmission.
- C) both direct contact and the respiratory route.
- D) penetration into unbroken skin.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

53) Meningitis can be caused by

- A) viral infections.
- B) bacterial infection.
- C) fungal infections.
- D) viral, bacterial, or fungal infections.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.5

54) Resident staphylococci are commonly found in the _____ and these seldom cause disease.

- A) eyes
- B) upper respiratory tract
- C) lungs
- D) kidneys

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

55) Loeffler's medium inhibits the growth of most organisms and is commonly used to cultivate

- A) *Corynebacterium diphtheriae*.
- B) *Mycobacterium tuberculosis*.
- C) *Mycobacterium leprae*.
- D) *Staphylococcus aureus*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

30.2 True/False Questions

1) The pink-red rash of scarlet fever may be caused by streptococcal pyrogenic exotoxins A, B, C, and F, which are coded for on a lysogenic bacteriophage.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

2) Infections by "flesh-eating bacteria" can occur when exotoxins A, B, C, and F and the bacterial surface M protein of *Streptococcus pyogenes* act as superantigens.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

3) The organisms that cause pneumococcal pneumonia are rarely found as a part of the normal flora of healthy adults.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

4) Reassortment of the genes of the influenza virus is facilitated by the fact that the genome is segmented.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.8

5) More than one type of influenza virus can infect a given cell at a given time.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.8

6) The prevalence of penicillin-resistant strains of *Neisseria gonorrhoeae* is low in North America and western Europe.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

7) A patient with AIDS often dies of a variety of infections caused by opportunistic microorganisms.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

8) The salt in mannitol salt agar inhibits the growth of *Staphylococcus aureus*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

9) most of the early cases of toxic shock syndrome in women were associated with use of highly absorbent tampons.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.9

10) The immune response may be as important as antibiotics in the elimination of *Bordetella Pertussis* from the body.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

11) The influenza viral envelope is unique because it contains only one type of protein.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.8

12) Chickenpox, caused by a herpes virus, results in a systemic papular rash that quickly heals.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.6

13) In an HIV infection the TH cells and macrophages are NOT commonly infected.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

14) A sudden onset of a headache, vomiting, and a stiff neck are all symptoms of meningococcal meningitis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.5

15) Viral hepatitis can result in acute kidney disease, followed by chronic kidney disease.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.11

16) Colds are the most infectious viral diseases.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.7

17) The availability of rapid antigen detection (RAD) systems for pathogenic streptococci has eliminated the need to culture *Streptococcus Pyogenes*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.2

18) Patients infected with *Mycobacterium Tuberculosis* in environments where drug resistance is expected or has already been identified usually have to take antibacterial drugs for six to nine months to ensure the infectious agent is completely killed.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.4

19) Most viral respiratory diseases are highly contagious but not life threatening.

Answer: TRUE

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 30.6

20) The loss of CD4 T-helper cells leads to the overt consequences of an HIV infection.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

30.3 Essay Questions

1) Explain the stages of syphilis. What are the characteristics of and treatments required at each stage?

Answer: The first stage of syphilis, called primary syphilis, begins with a localized infection and growth of *Treponema Pallidum* where a characteristic chancre lesion appears within two weeks to two months. Secondary syphilis occurs when the cells from the initial site of infection spread and a hypersensitive, generalized skin rash reaction occurs when the cells move into other locations such as mucous membranes, joints, eyes, bones, or the central nervous system. Both the first and second stages of syphilis are treatable by a single injection of benzathine penicillin G. Tertiary syphilis requires several injections over time of penicillin and has the most severe symptoms that can be bone and skin infections or, even worse, death caused by infections in the cardiovascular or central nervous systems.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

2) Explain the relationship between chickenpox and shingles, how each is contracted, and how each can be prevented.

Answer: Both chickenpox and shingles are caused by the same virus, varicella-zoster virus (VZV). Chickenpox is a common childhood disease caused by VZV, which is contracted via inhalation of close person-to-person contact. The VZV remains latent in nerve cells and when an individual becomes immunocompromised can cause the disease known as shingles. Vaccines are now available to prevent both of these diseases.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.6

3) Describe how the influenza vaccine is made in the United States. Explain the strain selection and vaccine production processes for the influenza vaccine.

Answer: Each year a new influenza vaccine is made by taking prevalent strains at the end of each influenza season and infecting embryonated eggs with the different strains, one at a time. The viruses are inactivated in the eggs, and all of the strain types are combined as a vaccine against influenza virus.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.8

4) Describe which diseases human papillomavirus (HPV) can cause in males and females. What is the most common symptom of HPV infection, and why is this of concern?

Answer: Females infected with HPV can develop cervical cancer, and men infected with HPV can develop anal and penile cancers. Individuals infected with HPV are most commonly asymptomatic, which is quite alarming considering the high number of those infected by the virus and its potential to cause cancer in both males and females.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.14

5) Explain the tuberculin test, including visual identification and immunological interpretation of positive and negative results, and describe why a positive test does not indicate an active infection.

Answer: The tuberculin test is a test performed to identify exposure to *Mycobacterium tuberculosis*, the causative agent of tuberculosis. The tuberculin skin test looks for hypersensitivity specific to tuberculin, a protein unique to *M. tuberculosis*. A local immune inflammatory response is elicited from tuberculin exposure to a hypersensitive individual within 1-3 days at the tuberculin-exposed area. This positive test result does not necessarily mean that the individual is currently infected by *M. tuberculosis*. A healthy person previously exposed to the pathogen could also show a positive test result, because the cell-mediated immunity is lifelong. A negative tuberculin test result is indicated by a lack of induration and edema at the intradermal injection site and suggests the individual has never been exposed to *M. tuberculosis*.
Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.4

6) Describe the disease caused by herpes simplex 1 virus (HSV-1) as well as how it is spread. Based on the disease characteristics, is HSV-1 capable of lysogeny?

Answer: The incubation time of HSV-1 is short (3-5 days) and the virus primarily infects epithelial cells around the mouth and lips, causing cold sores that heal without any treatment in 2-3 weeks. HSV-1 can be acquired either through direct contact or through saliva. After initial infection, HSV can become lysogenic in the nervous system, entering a lytic cycle periodically. The conditions that result in the virus becoming lytic are not well understood, but seem to be related to stress and immune system function.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.14

7) Describe the pathogenesis of HIV to explain how the number of CD4 lymphocytes decline in an ongoing HIV infection.

Answer: HIV commonly infects macrophages first, then targets TH cells. After HIV has infected macrophages, a different HIV gp120 protein is made that binds to the CXCR4 chemokine receptor on T cells. A T cell, once infected with HIV, can fuse with up to 50 other uninfected T cells, called syncytia, which results in the loss of immune function and eventually cell death.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15

8) Consider mumps. What is the causative agent? How is this disease spread? What happens in the body during the course of this disease? What type of immunity is conferred or used for this disease? Why is the prevalence of mumps in developed countries generally low?

Answer: Mumps are caused by a paramyxovirus. The virus is spread by airborne droplets and is highly infectious. The paramyxovirus spreads through the bloodstream (called viremia), which provides it access to infect and cause disease in organs such as the brain, testes, and pancreas. Antibodies are generally produced in response to this viral infection and usually lead to a quick recovery of the mumps disease. An attenuated mumps vaccine is commonly given to children in the form of the MMR vaccine and has been highly effective at preventing mumps, therefore the prevalence of mumps in developed countries is generally low.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.6

9) In the context of the influenza virus, describe the mechanistic difference between an antigenic drift and an antigenic shift. Also indicate the result of each.

Answer: Antigenic drift is a change in the influenza virus antigens due to a mutation within its genome. The mutations generally result in small changes in antigens, but an individual will not have immunity to the newly mutated antigen-containing influenza virus. When two or more different influenza virus strains rearrange the order of their genes (called reassortment), the influenza virus antigens tend to change in major ways. Again, an individual will not be immune to the newly re-assorted antigens.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.8

10) Explain the pathology, diagnosis, and treatment of gastric ulcers caused by *Helicobacter pylori*.

Answer: *H. pylori* can be spread either by person-to-person contact or by ingesting contaminated food or water. The pathogen colonizes the non-acidic gastric mucus layer of the stomach, where it produces a cytotoxin, lipopolysaccharide (LPS), and urease, which cause inflammation, tissue destruction, and ulceration. Chronic infection of *H. pylori* can lead to gastric ulcer formation. A clinical diagnostic such as the *in vivo* detection of urease by ingestion of ^{13}C urea and the presence of $^{13}\text{CO}_2$ likely indicates *H. pylori* presence, however identification of the pathogen from a gastric ulcer biopsy is most definitive. A two-week regimen combining an antibacterial compound (e.g., metranidazole), an antibiotic (e.g., tetracycline), and an antacid generally kill the pathogen.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.10

11) Why are most sexually transmitted diseases (STDs) transmitted only by intimate person-to-person contact?

Answer: Most STDs are sensitive to environmental stressors such as drying, heat, and light. The genitourinary tract where the pathogens thrive is protected against these stressors, so most are not able to adapt to other environments. Bodily fluids from the genitourinary tract are usually the only environment where they remain viable pathogens, so intimate contact is generally required to obtain STDs.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.13

12) Why do powerful antibiotics not always remove the cause of meningitis if the disease is caused by *Neisseria meningitidis*?

Answer: *N. meningitidis* is not the only causative agent for the disease meningitis. Other bacteria such as *Haemophilus influenza*, *Staphylococcus*, and *Streptococcus* can cause meningitis, but antibiotics would likely kill them. Meningitis can also be caused by viruses (called viral meningitis) such as enteroviruses, herpes simplex virus, lymphocytic choriomeningitis virus, and mumps virus, which would not respond to antibiotic treatment.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.5

13) Describe three antiretroviral drug types that are commonly used in highly active antiretroviral therapy (HAART).

Answer: Fusion inhibitors (e.g., enfuvirtide) bind to a membrane protein (gp41) that prevents the formation of the viral envelope of HIV and the CD4 cytoplasmic membrane of human immune cells. Integrase inhibitors bind to integrase enzymes and prevent HIV replication by halting the virus' ability to integrate its dsDNA genome into a host cell's DNA. Nucleoside reverse transcriptase inhibitors (NRTIs) stop the replication of HIV by terminating DNA synthesis due to the incorporation of a nucleoside analog (e.g., azidothymidine) that cannot form a bond with the next nucleotide in the HIV genome sequence when replicating. Non-nucleoside reverse transcriptase inhibitors (NNRTIs) also halt replication of the HIV genome, but NNRTIs directly interact with the reverse transcriptase enzyme and halt its function. Protease inhibitors (PIs) are peptide analogs that inhibit protein processing of HIV rather than nucleotide/genome replication, which prevent maturation of HIV.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.15